Do Now · retrieval

Boolean algebra rules

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Normalise 0.0110×2^4 using a two’s-complement mantissa. Give the new mantissa and exponent.

Q2

A sorted array is [3, 8, 12, 19, 27, 35, 42] . When searching for 27 , which indexes are checked, in order?

Q3

Show the array after one full left-to-right pass of ascending bubble sort on [7, 3, 5, 2] .

Q4

An alarm sounds if the door is open AND the system is armed, OR if panic is pressed. Write Q using A, B and C.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

De Morgan — CODE:
¬(A ∧ B) → ¬A ∨ ¬B
¬(A ∨ B) → ¬A ∧ ¬B — The operator inside the brackets swaps as each term is negated.

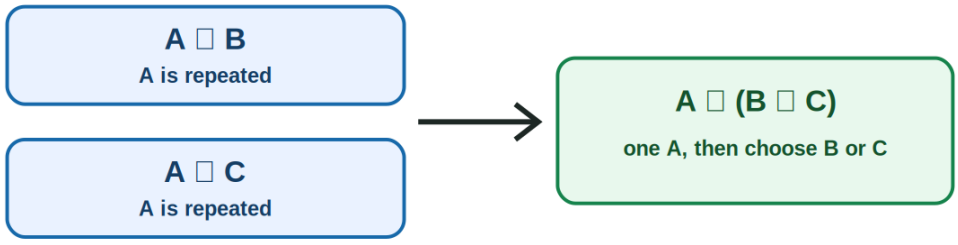
Core idea 2

Double negation — CODE:
¬(¬B) → B — Two consecutive NOT operations cancel.

Core idea 3

Distributive law — CODE:
(A ∧ B) ∨ (A ∧ C)
= A ∧ (B ∨ C) — A appears in both groups. Write it once outside the brackets, then join B and C with OR.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Association + commutation — CODE:
A ∨ (B ∨ C)
= (A ∨ B) ∨ C

A ∧ B
= B ∧ A — These allow regrouping and reordering before other simplification.

Vocabulary · use the correct technical terms

De Morgan — Rules for negating AND/OR groups.

distribution — Expanding/factoring Boolean expressions.

association — Regrouping repeated same operations.

commutation — Changing order.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Simplify ¬(A ∧ B)

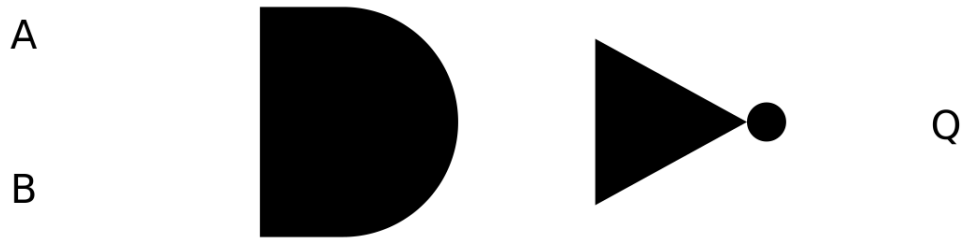
Follow the worked method

1. Outer NOT over AND — CODE:
Use De Morgan's Law.

2. AND becomes OR — CODE:
¬A ∨ ¬B

3. Equivalent behaviour — CODE:
Both are false only when A=1 and B=1.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the same expression
CODE:
¬A ∨ ¬B
= ¬(? ? ?)

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Simplify Boolean expressions

State the simplified form of each expression:

(i) ¬¬A (ii) ¬A ∧ ¬B (iii) ¬(¬A ∧ ¬B)

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

From a gate circuit to an expression

Two OR gates take inputs A, B and C, D respectively. Their outputs feed a third OR gate. Write the complete Boolean expression for the circuit output.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

De Morgan's Laws swap AND and OR while negating each term.

Use key vocabulary from memory

1. Explain De Morgan in your own words.

2. Explain distribution in your own words.

3. Explain association in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.